FINANCIAL SERVICES SECTOR DESIGN BASIS FRAMEWORK

Design Basis, Design Basis Accident, and Military-Action Extension Methodology
Function-Primary Facility-Class Architecture — a Consumer-Sector Framework

DBA-FIN: Financial Services Sector Parent Document

A standalone sector series in the Design Basis for AI Governance in Critical Infrastructure

Version 0.5 — DRAFT

October 2026

Tim Roxey
Eclectic Technologies

For review by engineers, regulators, financial-market-utility operators, and policy analysts responsible for the resilience of payment, settlement, clearing, and banking infrastructure.

SUGGESTED CITATION

T. Roxey, "DBA-FIN_Sector_Framework — Financial Services Sector Framework," Version 0.5, in CIKR Design Basis Examples, Edition 1 (November 2026), Eclectic Technologies, November 2026. https://doi.org/10.5281/zenodo.23863986

This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

Section 1 — Purpose, Scope, and Series Architecture

1.1 Purpose

This document establishes the Design Basis Framework for the National Infrastructure Protection Plan (NIPP) Financial Services sector. It is the parent instrument for the DBA-FIN series. It applies the DB → DBA → DBA-MA three-layer analytical progression to the financial functions whose continuity the economy depends on: payment and settlement, market infrastructure, and core banking.

1.2 What This Document Does Not Do

This document does not supersede financial regulation or supervisory guidance. It does not restate the Bright Line or Command Broker, inherited by reference from FD-BL. Critically, it does not re-scope the physical infrastructure the sector runs on: the data centers, network connectivity, and electric power that carry financial services are governed by the DBA-DC, DBA-COMM, and DBA-EN series respectively, and are inherited here rather than duplicated.

1.3 Function-Primary Organizing Principle

Placement follows the consequence chain. The Financial Services sector is organized here by financial function — the movement and settlement of value, the operation of markets, and the maintenance of accounts — rather than by institution type or by the physical assets the functions run on. Each function is placed by the consequence its failure produces, and is sub-indexed by systemic importance rather than by a physical mode or hazard class.

1.4 Series Placement — DBA-FIN as a Standalone Sector Series

DBA-FIN is a standalone sector series, a peer of the energy, water, chemical, communications, transportation, and nuclear series. It inherits the DB → DBA → DBA-MA methodology from the DBA-MA parent and the Bright Line and Command-Broker architecture from FD-BL, and it applies the mandatory Military-Action Extension Applicability Determination.

1.5 Financial Services as a Consumer Sector

Financial Services is the first consumer sector in the corpus, and its design basis has a different shape because of it. The sectors built so far — energy, water, chemical, communications, transportation — produce and move a physical thing, and their facility classes are physical facilities. Financial Services produces no physical thing; its infrastructure, as the CIKR taxonomy itself records, is a set of dependencies on data centers, telecommunications connectivity, and electric power. Its facility classes are therefore consequence-defined financial functions that run on inherited infrastructure, not new physical facility types.

This shape is deliberate and is the model for the consumer sectors that follow. DBA-FIN does not re-derive the design basis of the data center or the network; it inherits DBA-DC (including the WM-3 workload-migration and concentration analysis), DBA-COMM (connectivity and,

importantly, the position-navigation-timing overlay on which trade timestamping and settlement sequencing depend), and DBA-EN (power). What DBA-FIN adds — its novel and irreducible content — is the systemic-consequence layer: settlement finality, market continuity, and the cascade that the CIKR taxonomy labels economic paralysis and confidence erosion. The distinctive contribution of a consumer-sector framework is the consequence model, not the facility inventory.

Section 2 — The Financial Sector and Its Regulatory Context

2.1 A Multi-Regulator Sector with a Systemic-Designation Anchor

The Financial Services sector is supervised by a family of regulators — the Federal Reserve, the Office of the Comptroller of the Currency, the Securities and Exchange Commission, the Commodity Futures Trading Commission, the Federal Deposit Insurance Corporation, and, for systemic risk, the Treasury-chaired Financial Stability Oversight Council. Cyber and operational-resilience expectations are set through the FFIEC and implemented through the industry mechanisms FS-ISAC and Sheltered Harbor. Unlike the other sectors, Financial Services carries a built-in criticality designation.

2.2 The SIFMU Designation as the Criticality Anchor

The Financial Stability Oversight Council designates systemically important financial market utilities — the clearing houses, settlement systems, and payment systems whose failure would threaten the stability of the financial system. That designation is the financial-sector analog of the anchors the other frameworks use — the UN/DOT hazard class in chemical, the modal safety regime in transportation. It is an existing, authoritative systemic-importance taxonomy, and the DBA-FIN framework anchors its consequence tiering to it rather than constructing a bespoke one.

2.3 The Regulatory Gap

Operational-resilience guidance is comparatively mature in this sector — the FFIEC handbooks, the Sheltered Harbor data-vaulting standard, and supervisory stress testing all exist. What does not exist is a cross-industry engineering design basis stated in consequence terms: the set of events the financial system must be demonstrated to withstand, expressed as settlement-finality and market-continuity requirements rather than as institution-level controls. The systemic consequence is under-owned precisely because it spans institutions and regulators.

Section 3 — Function-Primary Taxonomy

3.1 The Function Classes and the Systemic-Importance Sub-Index

The sector is decomposed into three function classes — payment and settlement, market infrastructure, and core banking — each realized as a logical facility class, a system or service

running on the physical substrate this sector inherits, and one coordination class, the sector-coordination layer. Each is sub-indexed by systemic importance, keyed to the SIFMU and global-systemically-important designations, so that the design basis load a given instance carries scales with the systemic consequence of its failure.

3.2 Payment & Settlement (PS)

The payment-and-settlement function covers the systems that move value with finality: the Fedwire and CHIPS large-value systems, the ACH network, the card networks, and the emerging digital-payment platforms. Its governing consequence is loss of settlement finality within the operative window — intraday for large-value systems, T+1 for securities — and its defining property is that finality, once failed, cascades: a settlement that does not complete strands liquidity across every counterparty in the chain.

3.3 Market Infrastructure (MK)

The market-infrastructure function covers the securities exchanges, the central counterparties that clear trades, the DTCC settlement complex, and the derivatives markets. Its governing consequence is loss of market continuity — the ability to trade, clear, and settle in an orderly way — and its distinctive design-basis feature is the clearing-house default waterfall, whose failure converts a single-member default into a systemic event.

3.4 Core Banking (BK)

The core-banking function covers the deposit and account-ledger systems, the ATM networks, and the online and mobile banking platforms through which the public and businesses hold and move money. Its governing consequence is loss of account availability and cash distribution — the failures the CIKR taxonomy names as cash-distribution and payroll-processing failures, the point at which a financial-system disruption becomes visible to the public and erodes confidence.

3.5 OpsCo — Sector Coordination (coordination class)

The coordination function is carried by a coordination class spanning the industry mechanisms — FS-ISAC for information sharing and Sheltered Harbor for recovery data-vaulting — and the federal systemic-oversight function under the Treasury and the FSOC. On the coordination spectrum the corpus has developed, financial services sits near communications: it has a strong, partly mandated sector-coordination commons rather than the purely privatized coordination of the chemical sector. OpsCo — industry self-coordination through FS-ISAC and Sheltered Harbor under a systemic regulator (Treasury / FSOC) that can escalate to compelled action in a crisis — is the locus at which the AI-Governance Design Basis is implemented for the sector — the Bright Line placement and Command-Broker obligation for algorithmic and generative-AI systems in trading, clearing, fraud and anti-money-laundering, and automated settlement.

† OpsCo is introduced here as a coordination-class designation; the PS / MK / BK designators are ratified as of 2026-08-09. The consumer-sector shape — facility classes defined by financial function and inheriting their physical substrate from DBA-DC / DBA-COMM / DBA-EN — is ratified as the model for subsequent consumer sectors.

3.6 The Settlement-Finality Consequence Metric

The energy and communications series each established a governing consequence-duration metric. The financial-sector analog, proposed here for development, is the settlement-finality window: the continuous interval within which settlement finality must be achieved before the failure cascades into counterparty liquidity strain and, beyond a calibrated threshold, systemic instability. It is intraday for large-value payment and is bounded by the T+1 cycle for securities; its site-specific calibration and its Zone-based escalation structure are to be developed in the payment-and-settlement facility-class instrument. *Payment and Settlement* (DBA-FIN-PS-FC1) calibrates the window as five parameters, the finality window, recovery time, integrity verification time, extension capacity and stranded value, with three zones of escalation, and states the tension between resuming within the two hours the international principles aim at and resuming on verified records (*Payment and Settlement* §3.3, §10.2, §10.6); this framework takes the calibration up at its next conformance pass, after *Market Infrastructure* has been read against it.

Section 4 — The Financial Sector Design Basis

[Skeleton — to be developed.] The design envelope will be organized by event category with a consequence-severity inclusion criterion consistent with the series.

4.1 Design Basis Principles

The criterion for inclusion is systemic consequence severity — potential for loss of settlement finality, loss of market continuity, or loss of cash and account availability at a scale that produces the cascade the CIKR taxonomy names economic paralysis and confidence erosion — not event likelihood. Candidate bounding events, to be developed: settlement-system outage across the finality window; exchange or clearing-house halt and default-waterfall stress; core-banking and cash-distribution failure; and the inherited data-center, connectivity, and power failures that manifest as financial-service loss.

Section 5 — Dual Design Basis: Bright Line and Command Broker

[Skeleton — to be developed on the DBA-W §5.5 model.]

The financial sector is among the most advanced deployers of machine learning and generative AI — in algorithmic trading, real-time fraud and anti-money-laundering detection, credit decisioning, and increasingly in automated settlement and treasury operations. The Bright Line must be placed for any action of a component within scope of FD-LD §4.3 whose incorrect performance could move a market, settle a transaction irrevocably, or freeze account access at or above the systemic-consequence threshold; such actions require the architecturally enforced

authorization of a qualified Command Broker. Market infrastructure and the coordination class are the primary loci. Determinism decides scope under FD-LD §4.3; it does not decide placement.

Section 6 — Cross-Sector Dependencies and the Consequence Elevation Rule

6.1 Inward View — the Defining Dependencies

For a consumer sector the inward view is the defining one. Financial Services depends on the data-center series for its primary, disaster-recovery, and geographically-redundant processing; on the communications series for dedicated circuits and internet connectivity and, acutely, for the position-navigation-timing service on which trade timestamping and settlement sequencing rest; and on the energy series for the power that underlies all of it. A DBA-COMM PNT denial or a DBA-DC concentration event is, seen from this sector, a financial event, and these dependencies are stated as design basis conditions rather than assumed away.

6.2 Outward View — the Systemic Cascade

The outward consequence of Financial Services failure is economic rather than physical, but it is no less a cross-sector cascade: a settlement or payment failure interrupts payroll, cash distribution, and commercial transactions across every other sector, and beyond a threshold erodes the public confidence on which the monetary system rests. Financial Services is the substrate of economic activity, and its systemic failure is a Tier-3 consequence for the economy as a whole.

6.3 The Consumer-Sector Elevation — Reversed Direction

The consumer-sector shape reverses the usual elevation direction. In the producer sectors, a facility's failure elevates the design basis of the assets that depend on it. Here, because Financial Services runs on inherited infrastructure, the elevation runs the other way: a data-center or communications asset that is the sole substrate for a SIFMU-designated financial function inherits that function's systemic-importance tier, and must carry the tightened acceptance criteria the SIFMU designation implies. The forward-direction elevation obligation is therefore levied by DBA-FIN onto DBA-DC and DBA-COMM, not the reverse.

Section 7 — Military-Action Extension Applicability

[Skeleton.] Financial infrastructure is a primary objective of economic warfare, and the settlement backbone and the SIFMU-designated utilities are the highest-strategic-value nodes. Application remains a per-facility determination under the mandatory two-factor test — geolocation and strategic value — with strategic value driven by systemic-importance designation. Timing and settlement-sequence manipulation, and the denial of the PNT service on which they depend, are the distinctive financial-warfare vectors the MA extension must address.

Section 8 — Facility Class Register and Inheritance Model

8.1 Inheritance Model

Facility-class instruments of every type in the DBA-FIN series inherit the threat taxonomy, bounding methodology, consequence-analysis structure, adequacy standards, and Bright Line / Command-Broker architecture established in this framework, and — as a consumer sector — inherit the physical design basis of their substrate from DBA-DC, DBA-COMM, and DBA-EN. Each instrument adds the financial-function-specific consequence chain, the settlement-finality or market-continuity requirement, and the recovery architecture.

8.2 Facility Class Register

The following table is the current facility-class register for the DBA-FIN series. DBA-FIN-PS-FC1 has issued as a working draft and the other two logical classes are at Anticipated status; the physical substrate is shown as governed by the DBA-DC / DBA-COMM / DBA-EN series and inherited here. Designators are ratified as of 2026-08-09 and sub-indexed by systemic importance.

Designator	Function class (function-primary)	Governing characteristic	Status	Primary regulatory / designation anchor
DBA-FIN-PS-FC1	Payment & settlement systems — Fedwire, CHIPS, ACH, card networks, digital payment	settlement finality; intraday / T+1 window	Issued (working draft)	Federal Reserve; FSOC SIFMU designation
DBA-FIN-MK-FC2	Market infrastructure — securities exchanges, central counterparty (CCP) clearing, DTCC settlement, derivatives	market continuity; clearing-house default waterfall	Anticipated	SEC; CFTC; FSOC SIFMU designation
DBA-FIN-BK-FC3	Core banking & ledger — deposit/account systems, ATM networks, online & mobile banking	account availability; cash distribution	Anticipated	OCC; Federal Reserve; FDIC; NCUA
DBA-FIN-OpsCo†	Sector coordination (coordination class) — FS-ISAC, Sheltered Harbor, Treasury / FSOC oversight	systemic coordination & recovery	Anticipated (architecture)	Treasury / FSOC; FS-ISAC; FFIEC
(inherited)	Underlying substrate — data centers, network connectivity, electric power	governed by DBA-DC / DBA-COMM / DBA-ES	Governed elsewhere	see those series (consumer-sector inheritance)

Designators are ratified as of 2026-08-09. Facility-class numbers are sequential across the series; OpsCo is a coordination class and carries no FC number, and on the coordination spectrum it resolves to industry self-coordination under a systemic regulator (Treasury / FSOC) rather than mirroring DBA-ES-CntlCo. Each function class is sub-indexed by systemic-importance tier (SIFMU / G-SIB designation).

Section 9 — Development Status and Open Items

This is the initial working draft of the DBA-FIN sector framework and the first consumer-sector framework in the corpus. It establishes the standalone-series placement, the consumer-sector shape and its inheritance from DBA-DC / DBA-COMM / DBA-EN, the function-primary

decomposition (payment & settlement, market infrastructure, core banking, and the OpsCo coordination class) with the systemic-importance sub-index, the SIFMU designation as the criticality anchor, the proposed settlement-finality-window consequence metric, and the reversed-direction elevation onto the substrate series. The design-basis envelope (Section 4), the dual-DB detail (Section 5), and the MA applicability section (Section 7) are stubs to be developed. Open items at this version: develop the settlement-finality-window metric; and reconcile the reversed-direction elevation with the DBA-DC and DBA-COMM registers. At v0.5, DBA-FIN-PS-FC1 (*Payment and Settlement*) has issued as the series' first facility-class instrument and the second logical class written in the corpus, with large-value and wholesale settlement and the participant's interface to it in full and retail payment systems at the depth of the sub-index, and the three reference events it proposed are conferred at Section 9.1. It calibrates the settlement-finality window of Section 3.6 and develops the candidate settlement-system outage of Section 4.1. What it notes to this framework is taken up at the framework's next conformance pass: the five parameters and three zones of the window, the event categories it develops ahead of Section 4, the three tiers of the sub-index with the participant's endpoint carried with the system it reaches, the geolocation factor restated for a logical class, its declared residuals, the consequence-basis conditionality check, and a rule on resuming settlement on verified records (*Payment and Settlement* §10.10). Further open items: the home of the financial messaging network (*Payment and Settlement* §10.4); the question of what one action is when a component releases a payment into finality, which joins the question Project 0 holds from the information-technology series (*Payment and Settlement* §7.3, §10.5); and the three reference-event instruments of Section 9.1.

9.1 Conferred Reference Events

On the author's decision of 14 October 2026, this framework confers a reference-event designator on each of the three events its first facility class proposed. A designator commits the series to two things: a row in the corpus-wide DBA Reference Event Register, and a standalone reference-event instrument for that event. Under the determination of 9 October 2026 (*Reference Events Move to Their Own Project*), reference-event instruments are written and held by Project 11, DBA Reference Events; this framework keeps the conferral. None of the three instruments is yet written. The designators are numbered in the order the proposing class lists its events; the number records conferral, not issue. Each designator is sector-primary and single.

From *Payment and Settlement*, three events are conferred: DBA-FIN-RE-1, the theft from Bangladesh Bank's account at the Federal Reserve Bank of New York, February 2016; DBA-FIN-RE-2, the outage of the Bank of England's RTGS system, 20 October 2014; and DBA-FIN-RE-3, the operation of Ukraine's payment system through the war, from February 2022. The event in Ukraine is a continuing condition rather than a single day, and its instrument bounds it. Two events the plan for closing the seven sectors named for this sector are not conferred here, on the author's decision of the same date: the ransomware attack on the United States arm of the Industrial and Commercial Bank of China in November 2023 is held for *Market Infrastructure*

(DBA-FIN-MK-FC2), and the denial-of-service attacks on United States banks' online services of 2012 and 2013 are held for *Core Banking and Ledger* (DBA-FIN-BK-FC3), each to be proposed by that class when it is written. The three instruments are owed, and they are Project 11's to write. *Payment and Settlement* states the three as proposed at its v0.1 and conforms to the conferral at its next revision.

This framework is still a skeleton at Sections 4, 5 and 7. The three instruments read their events back through the design basis of *Payment and Settlement*, which develops it ahead of the framework, and what they return to the framework is taken up at its next conformance pass.

References

[To be completed.] Governing and inherited instruments: FD-BL and determination annexes; DBA-MA Parent Framework (IAEA L1); DBA Series Gaps and Recommendations (§1.7, MA applicability); DBA-DC (data-center substrate; WM-3; concentration); DBA-COMM (connectivity; PNT overlay); DBA-EN (power). Regulatory and coordination: Federal Reserve; OCC; SEC; CFTC; FDIC; Treasury / FSOC (SIFMU designations); FFIEC; FS-ISAC; Sheltered Harbor. Series instruments: DBA-FIN-PS-FC1 (*Payment and Settlement*), the series' first facility class; DBA-FIN-RE-1 to DBA-FIN-RE-3, the conferred reference events of Section 9.1, their instruments owed to Project 11. *Software, Hardware and Product Supply Chain* (DBA-IT-SW-FC3), whose structure the class follows and whose referred question it joins.

Revision History

0.5 October 2026 — T. Roxey. Section 9.1 is added and confers DBA-FIN-RE-1 to DBA-FIN-RE-3 on the three reference events that DBA-FIN-PS-FC1 (Payment and Settlement) proposed, on the author's decision of 14 October 2026: the Bangladesh Bank theft (February 2016), the Bank of England RTGS outage (20 October 2014) and Ukraine's payment system through the war (from February 2022). ICBC (November 2023) is held for Market Infrastructure and the 2012–13 denial-of-service attacks for Core Banking and Ledger. Each designator commits the series to a register row and a standalone instrument, none yet written; the instruments are Project 11's. The register at Section 8.2 enters the class as issued. Section 3.6 records the class's calibration of the settlement-finality window. Section 9 records the class's issue, lists what it notes to this framework for the next conformance pass, and records the messaging network and the referred action question as open. References add the class, the three designators and DBA-IT-SW-FC3. No change to the taxonomy, the designators of the classes, the sub-index or the design basis. Revision-history entries are left as records.

0.4 October 2026 — T. Roxey. Conformed the facility-class type of PS, MK and BK under the author's determination of 30 September 2026 (Facility-Class Types Are Physical, Coordination and Logical), on his ruling of 5 October 2026 carried by Project 0's handoff of that date. Section 3.1 called the three classes physical-function classes; they are logical classes, systems and services running on the inherited DBA-DC, DBA-COMM and DBA-EN substrate. Section 3.1

now names the function classes as the decomposition tier and types each facility class as logical, and Section 8.2 refers to the three logical classes; each site was read for its sense. Revision-history entries are left as records. No change to the design basis, the taxonomy, the designators or the register.

0.3 October 2026 — T. Roxey. Conformed to the October 2026 Edition of the Foundational Definitions on Project 0's work order of 3 October 2026. Section 5 states the scope of Bright Line placement by the determinism carve-out of FD-LD §4.3 rather than as any AI action, and states that determinism decides scope and not placement (P9-11, wording). Functional class is retired at every site under the author's determination of 30 September 2026, each site read for its sense: the decomposition tier reads function class, and the coordination layer reads coordination class (OI-15). No change to the design basis, the taxonomy or the register.

0.2 August 2026 — T. Roxey. Designators ratified; coordination class resolved to OpsCo (industry self-coordination under a systemic regulator); PS / MK / BK confirmed. Consumer-sector shape and reversed-elevation model ratified.

0.1 August 2026 — T. Roxey. Initial skeleton and first consumer-sector framework. Established DBA-FIN as a standalone sector series; introduced the consumer-sector shape (facility classes defined by financial function, physical substrate inherited from DBA-DC / DBA-COMM / DBA-EN, novel content in the systemic-consequence layer), checked against the CIKR mindmap, whose Financial "Infrastructure Dependencies" node confirms the consumer framing. Adopted the function-primary decomposition (Payment & Settlement / Market Infrastructure / Core Banking / OpsCo) with a systemic-importance sub-index anchored to the FSOC SIFMU designation; proposed the settlement-finality-window consequence metric; and stated the reversed-direction elevation onto the substrate series. Design-basis envelope, dual-DB detail, and MA applicability sections stubbed for development.